

Total No. of Questions : 8]

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**CE-227-CBCS**

**B.E. IV Semester**

Examination, June 2020

**Choice Based Credit System (CBCS)**

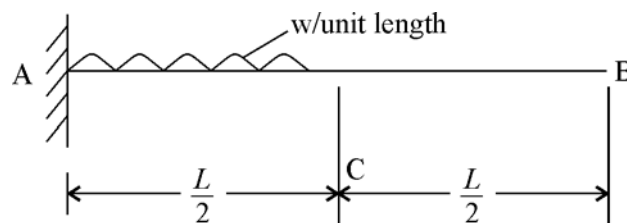
**Structure Analysis - I**

**Time : Three Hours**

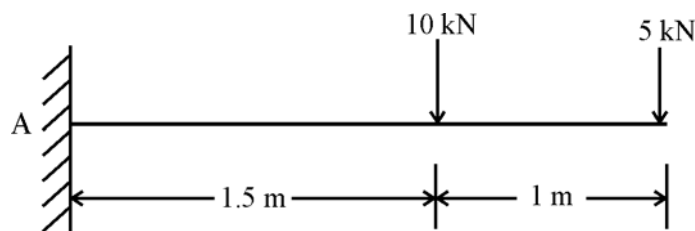
**Maximum Marks : 60**

- Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.  
iii) Assume data suitably.

1. a) Find the rotation and deflection at the free end in the cantilever beam shown in fig.



- b) Determine the slope and deflection at the free end of a cantilever beam as show in figure by moment area method.

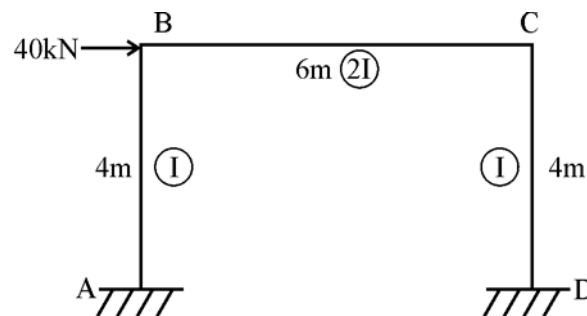


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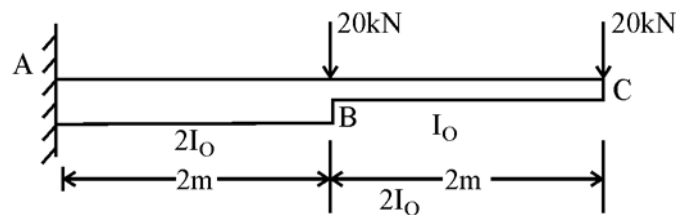
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2. a) Distinguish between static indeterminacy and kinematic indeterminacy.
- b) Analyze the frame shown in figure by moment distribution method and draw the bending moment diagram.



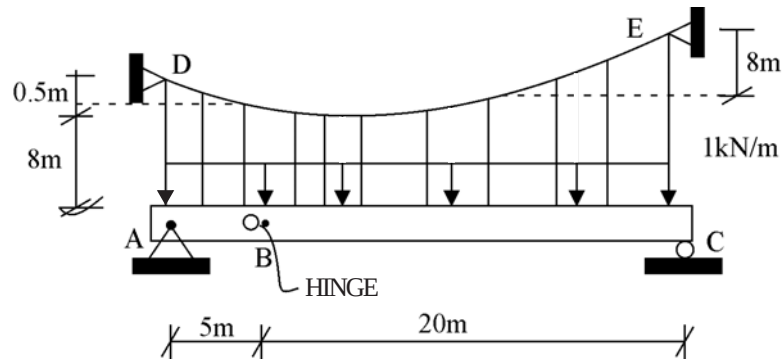
3. a) State the assumption made in slope deflection method for the analysis of indeterminate structures.
- b) Determine the deflection and rotation at the free end of the cantilever beam in figure. Use unit load method.  
Given  $E = 2 \times 10^5 \text{ N/mm}^2$  and  $I = 12 \times 10^6 \text{ mm}^4$



4. For the structure shown:
  - a) Determine the maximum tension of the cable.

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- b) Draw quantitative shear and bending moment diagram of the beam.



5. A single rolling load of 100kN moves on a girder of span 20m.
  - a) Construct the influence lines for
    - i) Shear force
    - ii) Bending moment for a section of 5m from left support
  - b) Construct the influence line for point at which the maximum shears and maximum bending moment develop. Determine the values.
6.
  - a) What is an Arch? Explain the differences between basic action of an arch and a suspension cable.
  - b) Define continuous beam. What are its advantages over simply supported beam?
7.
  - a) Explain the difference between Real work and virtual work. Also explain the principle of virtual work for deformable bodies.
  - b) A 3 hinged arch of span 40m and rise 8m carries concentrated loads of 200kN and 150kN at distance of 8m and 16m from the left end and an UDL of 50kN/m on the right half of the span. Find the horizontal thrust.
8.
  - a) Explain eddy's theorem with proper explanation.
  - b) What are Influence lines? What are the uses of influence line diagram?

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